

Accurate Angle Representation from Misplaced Hall-Effect Switch Sensors for Low-Cost Electric Vehicle Applications

Carlos Miguel-Espinar*, Daniel Heredero-Peris*, Gabriel Igor-Gross*, Marc Llonch-Masachs*,
and Daniel Montesinos-Miracle*, *Senior Member, IEEE* * Centre d'Innovació Tecnològica en Convertidors
Estàtics i Accionaments (CITCEA), Departament d'Enginyeria Elèctrica,
ETS d'Enginyeria Industrial de Barcelona, Universitat Politècnica de Catalunya, 08028 Barcelona, Spain
Email: carlos.miguel@upc.edu

Abstract—This paper presents an accurate method to determine the mandatory rotor position to control a Permanent Magnet Synchronous Machine (PMSM) from misplaced Hall-Effect Switch Sensors (HESS)s. The main feature of this technique is the estimated speed from two consecutive edge transitions in the same phase to dodge the inaccurate information. Besides, when a new electrical cycle starts, a gradual compensation algorithm corrects the initial speed estimation to cancel the electrical angle error in order to prevent current distortions. Furthermore, a First-Order Taylor (FOT) series approximation estimates the rotor position from the compensated speed calculation, reducing the possible torque ripple and noise due to an inaccurate control strategy. The proposed method has been compared with other state-of-the-art approaches through simulation and experimental results obtained from a low-voltage powertrain, showing more straightforward implementation and delivering the best balance performance during steady-state and transient operation.

Index Terms—First-Order Taylor Series, Hall-Effect Switch Sensor, Permanent Magnet Synchronous Machine, Positioning Error, Speed and Position Estimation.

I. INTRODUCTION

IN the past few years, the use of micromobility, such as e-scooters, e-bikes or e-motorbikes, in congested urban areas has undergone enormous growth [1]. The majority of these vehicles have a Permanent Magnet Synchronous Machine (PMSM) as main traction component because of its high power density, precise control and maximum torque at low speed, easy maintenance, reduced size, and high-efficiency [2], [3]. Several torque-and-speed regulation schemes are found in the scientific literature, such as Direct Torque Control (DTC) [4] or V/f control [5]. However, the most used and widely accepted in the industry sector is the Field Oriented Control (FOC) algorithm due to its highest robustness and independence from sudden motor parameter changes or external disturbances [6]. This strategy requires precise and continuous knowledge of the rotor position, which is mandatory to generate smooth and constant torque and reduce noise and vibrations. It is directly translated into a more extended lifetime and better performance of the entire powertrain.

The rotor position can be obtained from a mechanical or electromagnetic sensor or through a sensorless method.

Sensorless methods are classified as Back Electromotive Force Method (BEFM) [7]–[9] and High-Frequency Injection Method (HFIM) [10]–[12]. On the one hand, the BEFM strategy presents main problems with the starting performance and a rough position estimation at low speed [7], and it is hard to achieve a stable speed response with low loads [9]. On the other hand, the main drawbacks of the HFIM strategy are: i) requirement of a consistent portion of the DC bus voltage, ii) sensitivity to different load conditions, iii) difficulty to use with motors with low saliency ratio, and iv) influence of motor parameter variations, motor temperatures, and non-linear characteristics of inverters.

For electric vehicle (EV) applications, mechanical or electromagnetic sensors are used to get the proper rotor position: encoders, resolvers, and Hall-Effect Switch Sensors (HESSs). On the one hand, optical encoders are expensive, mechanically fragile, and susceptible to noise. Moreover, they demand large space [13]. Besides, incremental encoders have high-resolution, but an additional algorithm is necessary to obtain the correct position during the start-up [14]. On the other hand, resolvers have high-resolution and guarantee field orientation but increase the cost of the machine and require extra mechanical structure [15], [16].

On the contrary, HESSs are suitable for low-cost, mass-produced applications; for instance, micromobility or home appliances, despite the fact that they have several drawbacks [17], [18]: i) low resolution ($\pm 30^\circ$ E), ii) misplaced sensors due to incorrect installation, iii) intrinsic hysteresis, and iv) non-regular electrical periods due to differences in the magnetic field produced by adjacent magnets. These problems will affect the current waveform and the output torque, as well as the general performance of the machine, resulting in extra losses caused by the generation of harmonics that will overheat the electric machine after a long time operation [19]. The literature offers several methodologies to transform the discrete information of the three HESSs located in the electric machine, ideally separated (120° E) and dividing an electrical turn into six different sectors, into a continuous representation. Hence, the electrical rotor position and speed are estimated through an accurate mathematical algorithm. As per the authors' best knowledge, the literature shows several algorithms that focus on solving the misplaced HESS problem [20]–[33]. They are

briefly detailed in the following lines.

Morimoto *et al.* [20] applied both a Zeroth-Order Taylor (ZOT) and a First-Order Taylor (FOT) approximations to obtain the machine speed and the electrical angle, respectively. They considered a constant speed in each electrical sector and equal to the average velocity of the previous sector. The main benefit of this method is the low execution time and low memory usage on a microcontroller. However, this is not suitable if the reference speed radically changes, e.g. during accelerations and deceleration. Furthermore, it is susceptible when misplaced HESSs and not regular magnets are considered.

Capponi *et al.* [21] and Ozturk *et al.* [22] presented an FOT and a Second-Order Taylor (SOT) approximations to estimate the speed and electrical angle, respectively. Then, the speed and position are tracked more precisely compared with the algorithm introduced in [20]. Nevertheless, slight differences in the angular width of each sector cause variations in the estimated acceleration even at constant speeds. The acceleration estimate will turn from negative to positive values, provoking torque pulsations and noise.

In [23], [24], the authors presented a moving average algorithm to estimate the speed. Zhang *et al.* [23] obtained the rotor position through a linear interpolation of the previous electrical sector, based on a Least Square Method (LSM), to calculate the speed and initial electrical angle of the next electrical sector. Besides, a compensation algorithm at each Hall edge pulse is considered to correct the inaccurate electrical angle representation. The main drawbacks of this algorithm are its high execution time, memory usage, and discontinuity in the electrical speed at each Hall transition that provokes bumps and degrade the control operation. Kreindler *et al.* [24] calculated the speed through a moving average of six samples, and an SOT approximation is used to obtain the electrical angle. However, the estimation of the acceleration can be affected by noise, and the electrical angle is limited at its maximum values in each sector, not considering the misplacement of HESSs, which implies a current distortion.

In [26]–[32], a vector-tracking phase-locked loop method with the assistance of Hall-effect sensors based on torque estimation, back-electromotive estimation, power closed-loop, or cross-production phase detection are presented. Yoo *et al.* [26] and Wang *et al.* [27] deployed a dual observer and a position sensor offset compensation strategy to estimate the rotor speed and position without time delay or bumps. Nevertheless, the torque observer requires highly complex calculations and is deeply dependent on the motor parameters. In [28], [29], the electrical speed and position are calculated through a phase-locked loop structure to compensate for the misalignment of HESSs and to enhance the transient operating capability. Nevertheless, it is not suitable for constant torque at low speed. Moreover, the observer design is too complex and, again, is highly dependent on the motor parameters. Yu *et al.* [30] proposed a strategy in which the speed is measured through a ZOT approximation, and the position is based on the power closed-loop. This algorithm attempts to compensate for the Hall sensor position error and to estimate the rotor position continuously and accurately. The main drawbacks are that

the accuracy in determining the electric machine parameters entails a lot of experimental tests, and only copper losses are considered. In [31], [32], the combination of the HESS along with an orthogonal phase-locked loop is shown. The small position error and the dynamic performance are its main advantages. However, the main disadvantages are that variable electrical frequency implies variable cut-off frequency of low-pass filters and adaptive controller parameters.

The authors proposed an algorithm to calculate the rotor speed and position to lessen the consequences of having misplaced HESSs [33]. This methodology differentiates from state-of-the-art methods in terms of simplicity and generally better performance during transient conditions and steady-state operation. The algorithm is based on calculating the electrical speed from two straight edges in the same phase to avoid the misplaced position of the HESSs. Besides, a compensation algorithm is applied to the speed value to cancel the electrical angle error when a new electrical cycle starts. Finally, an FOT approximation estimates the rotor position based on the electrical speed estimation.

This paper broadens the study done in [33] by comparing the proposed methodology with other approaches through simulations and experimental results. The article is organized as follows: Section II explains the effect of misplaced HESSs. Section III presents the proposed method to estimate the speed and the electrical angle to avoid the misplacement of HESSs. Several simulation and experimental results claim the effectiveness of the proposed method in Section IV and V, respectively. Finally, Section VI concludes this article.

II. MISPLACED HALL-EFFECT SWITCH SENSORS

A. Ideal Hall signal without offset

HESSs provide the electrical rotor position at every 60° E, as long as these are precisely separated 120° E from each other. This information is constantly repeated pair of poles times along a mechanical turn, assuming that the permanent magnets are well-distributed around the rotor.

Due to some limitations in the technology to assemble the stator lamination sheets with the Hall sensors in all number of slots per pole combinations, it cannot guarantee the precise position of the sensors in some cases. Moreover, for electric machines with a large number of magnetic poles, the low-mechanical displacement between the ideal and actual place of HESSs translates into a more significant error in electrical degrees [17]. Hence, in a real machine, the HESS edges do not have to be equally spaced 60° E, leading to significant computational errors in the estimated speed and electrical position. As a result, the control performance will degrade, increasing the harmonic contents of current, torque oscillations and poor control dynamics.

Fig. 1 shows a cross-sectional view of an equivalent two-pole PMSM with ideal position of HESSs. In this example, each winding and Hall sensor phase is well-located, and no phase-shift is observed. So, the base defined by the physics of the machine and the base defined by the HESS information, which is used to control the machine currents generated by the traction inverter, are well-aligned.

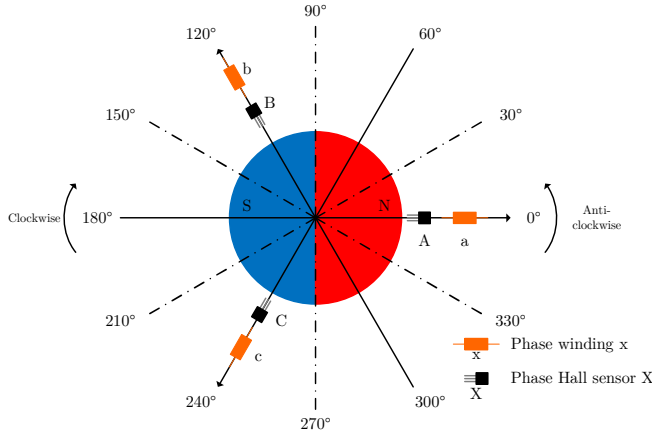


Fig. 1. Ideal spatial layout of phase windings and HESSs.

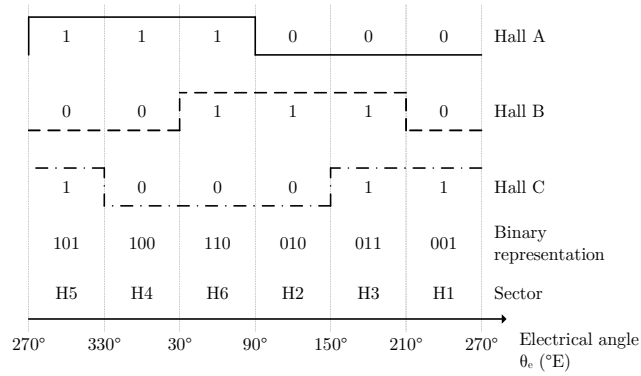


Fig. 2. Time diagram representation of HESSs in anti-clockwise direction, binary and numerical representation of the electrical sector, and theoretical electrical angle at each edge.

Fig. 2 displays the timing diagram of HESSs (solid, dashed, and dot-dashed lines for phases A, B, and C, respectively) if the machine turns in an anti-clockwise direction. The output signal of a HESS is electrically defined as high-level or low-level that is digitally represented as 1 or 0 respectively and depends on the rotor position. The three signals that are gathered from each phase set up a binary representation of the electrical sector. The sector number is calculated by considering the most and the least significant bit from phases A and C, respectively. Two straight edges are separated 60° E, as long as no spatial offset is considered between a HESS and its corresponding winding.

If the machine is spinning and the traction inverter detects entering a new electrical sector, the electrical angle (θ_e) will be updated depending on the angle representation method. At low speed, generally, the electrical angle is represented discretely following a 120° E commutation technique, which is used usually to control a Brushless DC (BLDC) Motor [17], [18]. Nevertheless, the electrical angle is continuously represented at high speed to avoid current ripple and diminish noise and vibrations.

Table I shows the electrical angle assignment at Hall sector ($\theta_{e,Hall(z)}$) depending on the current position of HESSs. During discrete representation, the mid-angle of each sector is set.

TABLE I
ELECTRICAL ANGLE SETTING

Hall Sector Hall(z)	Discrete $\theta_{e,Hall(z)}$ (°E)	Continuous	
		Clockwise $\theta_{e,Hall(z)}$ (°E)	Anti-clockwise $\theta_{e,Hall(z)}$ (°E)
1	240	270	210
2	120	150	90
3	180	210	150
4	0	30	330
5	300	330	270
6	60	90	30

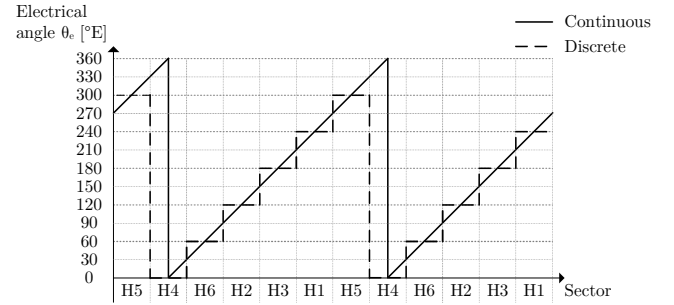


Fig. 3. Continuous and discrete representation of the electrical angle if the electric machine is spinning in an anti-clockwise direction and no misplacement of HESSs is considered.

This strategy leads to minimum angle-error and torque-ripple over a certain electrical sector. Nevertheless, during continuous representation, the updated angle at the Hall edge depends on the turning direction.

Fig. 3 displays the electrical angle representation regarding the time that is used to synchronise the traction inverter with the electric machine if no misplacement of the HESSs is considered. The dashed line represents the discrete information from the HESS used to control PMSMs at low speeds, whereas the solid line represents the continuous representation of the electrical angle used to control PMSMs at high speeds. The primary goal of this paper is to obtain the constant representation of the electrical angle when HESS misplacement is considered. It will avoid the current harmonics generated by the traction inverter due to an inaccurate representation of the electrical angle.

B. Consequences of Hall signals' misplacement

The displacement between the rotor position and the information from HESSs is mandatory for a traction inverter to perform torque and speed control with adjustable current phase lead angles. Each Hall sensor can be misplaced at a certain mechanical angle regarding its corresponding theoretical phase-winding position because of mechanical errors in the manufacturing or assembling processes. Tsotoulidis *et al.* [18], mention the two major mounting configurations for Hall sensors (mounted inside stator slots or mounted in an external printed circuit board outside the motor) and indicates how in low-cost applications, the Hall sensors are not precisely placed 120° E apart, showing their side effects in the machine performance. These errors would have consequences

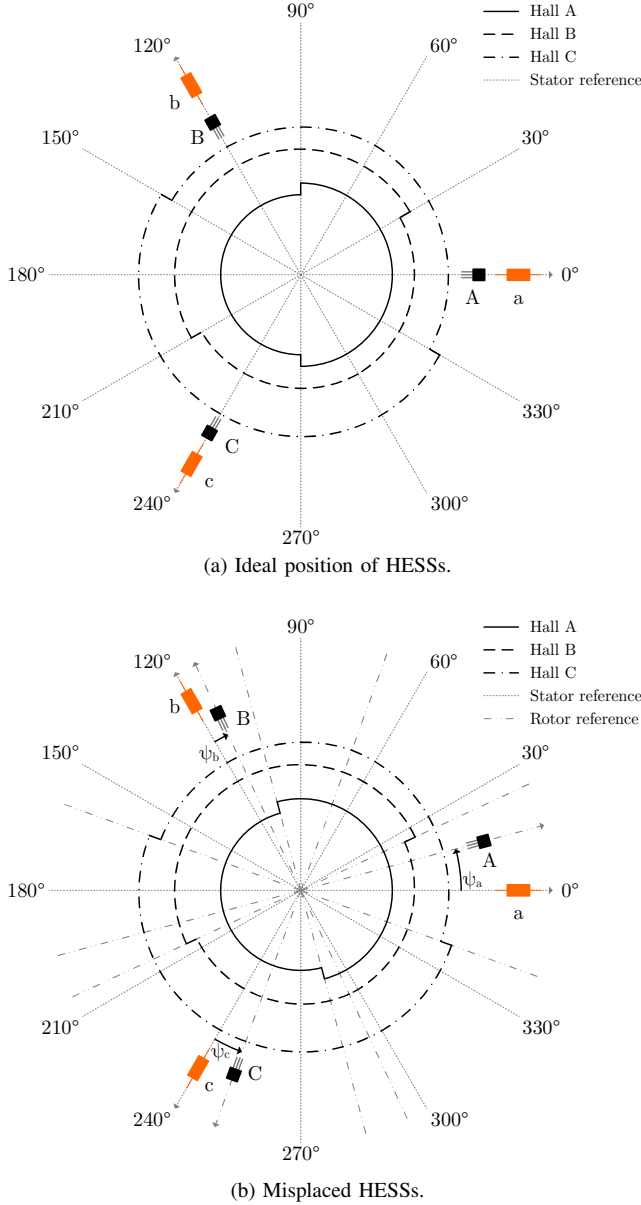


Fig. 4. Circle diagram of Hall signals in one electrical period with (a) ideal position of HESSs and (b) misplaced HESSs.

in miscalculating the speed and electrical angle. Therefore, the performance of the torque and speed controllers will be altered, deriving in vibrations, noise, and unstable controllability, and leading to the deterioration of the overall inverter performance.

Fig. 4 depicts the cross-sectional view of an equivalent two-pole PMSM with the spatial distribution of Hall sensor signals. It compares the electrical base formed from the ideal position of HESSs (see Fig. 4a) and considering their possible misplacement (see Fig. 4b). The densely-dashed lines refer to the physical machine base. The dot-dot-dashed lines indicate the rotor base created from the Hall sensor information. Solid, dashed, and dot-dashed lines illustrate the Hall sensor signals for phases A, B, and C, respectively.

An incorrect positioning of the three Hall sensors with unequal misplacement angle, ψ_a , ψ_b , and ψ_c has been considered

TABLE II
SHIFTED ELECTRICAL ANGLE SETTING

Hall Sector Hall(z)	Continuous	
	Clockwise $\theta_{e,Hall(z)} (^{\circ}E)$	Anti-clockwise $\theta_{e,Hall(z)} (^{\circ}E)$
1	$270 + \psi_a$	$210 + \psi_b$
2	$150 + \psi_c$	$90 + \psi_a$
3	$210 + \psi_b$	$150 + \psi_c$
4	$30 + \psi_b$	$330 + \psi_c$
5	$330 + \psi_c$	$270 + \psi_a$
6	$90 + \psi_a$	$30 + \psi_b$

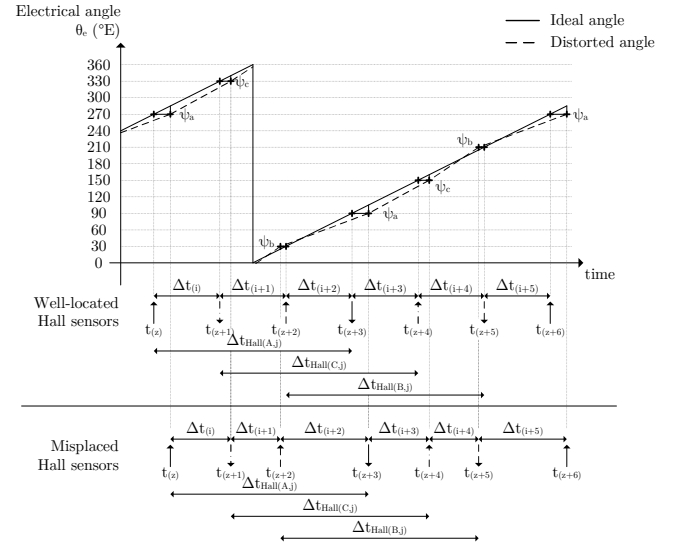


Fig. 5. Hall sensor edges with well-located and misplaced sensors.

respectively, concerning the winding of phases a, b, and c. Hence, TABLE II shows the corrected value to each Hall sensor edge that is different from the theoretical angle shown in TABLE I and the adjustment to be done which depends on the spinning direction.

Fig. 5 depicts the rising and falling edges in both the case of well-located and misplaced Hall sensors. The solid line exemplifies the ideal continuous representation of the electrical angle if no misplacement is esteemed, whereas the dashed line is the actual representation if the HESSs are misplaced. As exemplified by the dashed line, the electrical angle representation is distorted by the imprecise position of the HESSs. An advanced algorithm is required to smooth away this distortion to avoid generating harmonic currents to the electric machine translated to low-frequency torque ripple, noise and vibrations.

III. ELECTRICAL SPEED AND ANGLE CALCULATION

A. Speed estimation from HESS signals

In case the HESSs are well-located and the machine rotates at a constant speed, the formula to calculate the speed from the time difference between two consecutive edges is defined as

$$\omega_{e(i)} = \frac{\pi}{3} \frac{1}{\Delta t_{(i-1)}} \quad \forall i \geq 2, \quad (1)$$

where $\omega_{e(i)}$ is the electrical speed predicted for the sector i , and $\Delta t_{(i-1)}$ is the elapsed time between two consecutive edges during the sector $i-1$. Equation (1) commits considerable error with misaligned sensors affecting the electrical angle representation and, as a consequence, the generated current waveform. The time difference between two consecutive edges may be higher or smaller than the theoretical value at constant speed, as can be seen in Fig. 5.

This paper proposes a mathematical algorithm based on a moving average of three samples to calculate the electrical speed. In this specific way, the speed and electrical angle estimation will not be affected by the misplacement of HESSs. From Fig. 5 it can be observed that the elapsed time between two consecutive edges from the same phase ($\Delta t_{Hall(X,j)}$) is equal to

$$\Delta t_{Hall(X,j)} = \Delta t_{(i)} + \Delta t_{(i+1)} + \Delta t_{(i+2)} \quad \forall X = A, B, \text{ or } C, \quad (2)$$

$$\forall j \geq 1, \forall i \geq 1,$$

where the subscript X refers to the phase A, B, or C of HESSs, the subscript j indicates the number of periods of the same phase, and the subscript i is equal to $3j-2$.

By analysing Fig. 5 in detail, it can be demonstrated that the elapsed time between two consecutive edges from the same phase is not altered by the misplaced position of HESSs, as long as the permanent magnets are the same size, and they are evenly distributed around the rotor core. Hence, this paper proposes to estimate the electrical speed as

$$\omega_{e(i)} = \frac{\pi}{\Delta t_{(i-1)} + \Delta t_{(i-2)} + \Delta t_{(i-3)}} \quad \forall i \geq 4. \quad (3)$$

B. Speed compensation and electrical angle estimate

Suppose the electric machine runs at low speed, and the 120° E commutation strategy is applied. In that case, the assigned electrical angle, when an HESS is detected, is the same as indicated in Table I for the discrete column, depending on the electrical sector in which the rotor is located. When more than six edges have been detected, an entire electrical turn has been completed, and the electrical angle is continuously calculated according to an FOT series. At constant movement, the average speed of each electrical sector is considered steady, and it is calculated from the previous mid-electrical period following (3). The continuous angle representation would be computed as

$$\theta_{e(k)} = \theta_{e(k-1)} + \omega_{e(i)} \cdot (t_{(k)} - t_{(k-1)}) \quad \forall k \geq 1, \forall i \geq 4; \quad (4)$$

where, $\theta_{e(k)}$ and $\theta_{e(k-1)}$ are the electrical angles at the discrete instants k and $k-1$, respectively, and $t_{(k)} - t_{(k-1)}$ is the sample period at which the electrical angle is calculated.

During accelerations or decelerations of the machine, the estimated electrical position of the rotor may be different from (4). Some authors have considered updating the misleading angle immediately at each Hall sensor edge [21], [22]. Nonetheless, this strategy provokes an instantaneous change in the electrical angle translated into an abrupt change in the current waveform and, therefore, in the applied torque. Other authors have proposed to distribute the actual error along the time of the following electrical sector [24]. This approach

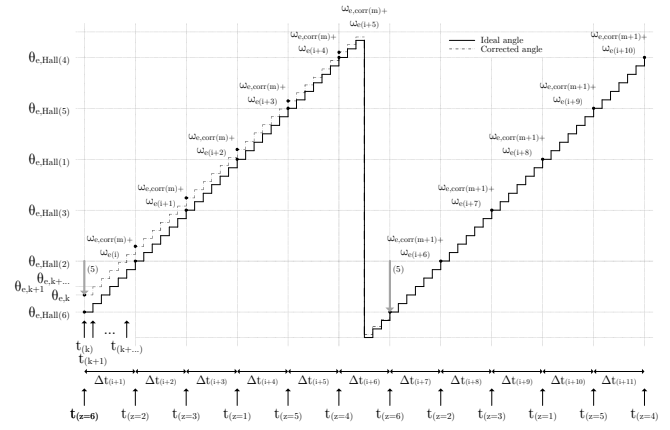


Fig. 6. Qualitative discrete implementation of the electrical angle estimation.

prevents instantaneous changes in the angle representation, but it continuously modifies the electrical speed in each electrical period provoking low-frequency distortions.

This paper aims to correct the electrical speed once in each electrical period. In the clockwise direction, the electrical speed is corrected at H5, whereas in the anti-clockwise direction is corrected at H6. Hence, an average speed correction ($\omega_{e,corr(m)}$) is applied during the entire m electrical period at the beginning of it as

$$\omega_{e,corr(m)} = \frac{\theta_{e,Hall(z)} - \theta_{e(k)}}{\Delta t_{(i-1)} + \Delta t_{(i-2)} + \Delta t_{(i-3)}} \cdot \frac{1}{2} \quad (5)$$

$$\forall Hall(z) = 5|6, \forall m > 1, \forall i \geq 4.$$

So, considering non-constant speed, the electrical angle can finally be estimated as

$$\theta_{e(k)} = \theta_{e(k-1)} + (\omega_{e(i)} + \omega_{e,corr(m)}) \cdot (t_{(k)} - t_{(k-1)}) \quad (6)$$

$$\forall k \geq 1, \forall i \geq 4, \forall m > 1,$$

also achieving no deviation between the HESSs and the stator windings at the steady-state due to the gradual assignment of $\theta_{e,Hall(z)}$ shown in Table II. Fig. 6 illustrates the discrete implementation of electrical angle according to (6), assuming the average correction of the electrical speed at H6 in the anti-clockwise direction according to (5).

In summary, this paper proposes a procedure to estimate the speed and electrical position of the rotor based on three different steps in an actual microcontroller, as seen in Fig. 7. First, the average electrical speed is calculated whenever a new Hall sensor edge is detected (at six times higher the electrical frequency) and at least two straight edges from the same phase have occurred, using (3). This calculation is done at each Enhanced Capture (ECAP) interruption. Secondly, the electrical speed is corrected at the beginning of an electrical period from (5). This task is done at each six HESS edges in the ECAP interruption. Finally, the electrical angle is updated, based on (4) when the speed correction is null or (6) when the speed correction is different from zero. This last task is implemented in a periodic timer interruption that is generally executed at the frequency of low-level control loops.

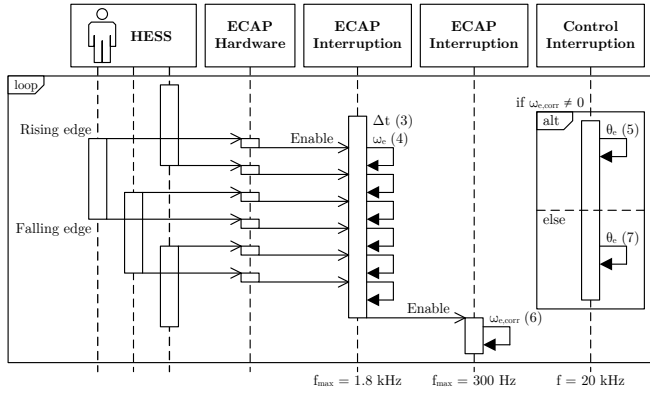


Fig. 7. Diagram of the sequence of speed and angle estimations.

TABLE III
IPMSM CHARACTERISTIC PARAMETERS

Symbol	Description	Value	Unit
R_s	Winding phase resistance	17	mΩ
L_d	d-axis inductance	70	μH
L_q	q-axis inductance	79	μH
λ_m	Flux linkage	0.016	(V·s)/rad
V_{DC}	Nominal voltage	48	V
P_n	Nominal power	2.5	kW
$\omega_{m,n}$	Nominal speed	500	rpm
$\omega_{m,max}$	Maximum speed	900	rpm
pp	Pair of poles	20	
ψ_a, ψ_b, ψ_c	Phase-shift angle	5.9, 6.5, 4.5	°E

IV. SIMULATION RESULTS

Simulations from a MATLAB-Simulink model are carried out to compare the accomplishment of the proposed method with its state-of-the-art counterparts. A direct drive Interior Permanent Magnet Synchronous Machine (IPMSM), which runs as the rear wheel of an e-motorbike, is used. In this article, an offline test has been done to measure the phase-shift angle between the flux-linkage and the HESSs signals to avoid any deviation between the stator windings and the HESSs. The most relevant parameters of the electric machine are listed in Table III.

Fig. 8 shows the control strategy. It is based on a Maximum Torque per Ampere (MTPA) algorithm for speeds below the rated speed and a Flux Weakening (FW) technique from the nominal speed up to the maximum achievable speed, as proposed in [34]. The block where the speed and electrical position are calculated is remarked with a dashed line. The subsystems directly affected by these variables are remarked with dash-dotted lines: i) Park and Inverse Park transformation and ii) current and speed PID controllers. Since the machine has a sinusoidal winding, i.e. the back electromotive force is sinusoidal, current harmonics will only appear due to an incorrect speed and electrical angle estimation.

This paper compares different methods to estimate the speed and electrical angle from misplaced HESSs: Method 1 [20], Method 2 [23], and Method 3 as the proposed method. Method 1 uses an FOT algorithm to calculate the electrical angle and a ZOT-based method to estimate the electrical speed based on the last two consecutive edges. Moreover, the electrical angle

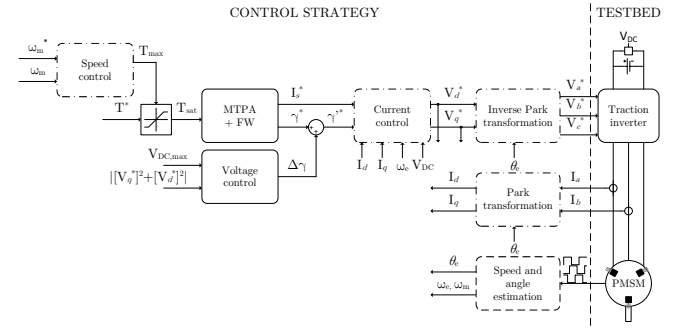


Fig. 8. Control algorithm strategy.

is assigned to the theoretical value at each HESS edge. Method 2 uses an FOT algorithm to calculate the electrical angle and a linear interpolation based on the Least Square Method to calculate the average speed and the initial angle of the next electrical period. Furthermore, the electrical angle is corrected through a gradually adjusted method at each HESS edge.

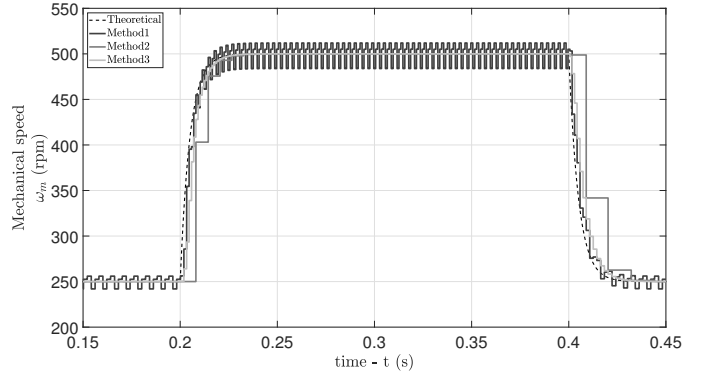


Fig. 9. Estimated mechanical speed during a speed transient if the electric machine runs as a generator.

Fig. 9 and Fig. 10 show the transient response of the mechanical speed, the electrical angle, and the electrical angle error when the electric machine runs as a generator and the speed increases from 50% of its nominal value up to 100% at time 0.2 s and goes down from 100% to 50% at time 0.4 s. During both speed steps, the precise evolution of the electrical angle representation is shown in Fig. 10b and Fig. 10c. Method 1 shows a quicker response and minor error in the electrical angle transient because it immediately corrects the deviation between the current value of the electrical angle and the actual value corresponding to the Hall event, whereas Method 2 and 3 averagely correct this deviation in the following electrical period. However, Method 2 minimizes, and Method 3 nullifies the misplacement of HESSs in the speed representation during the steady-state operation. Fig. 10d reveals difference between the theoretical and estimated electrical angle at steady-state for all methods. This error comes from the discrete implementation of the FOT approximation [13], [24].

Fig. 11 exhibits the steady-state performance of the traction inverter when the electric machine runs as a motor at nominal

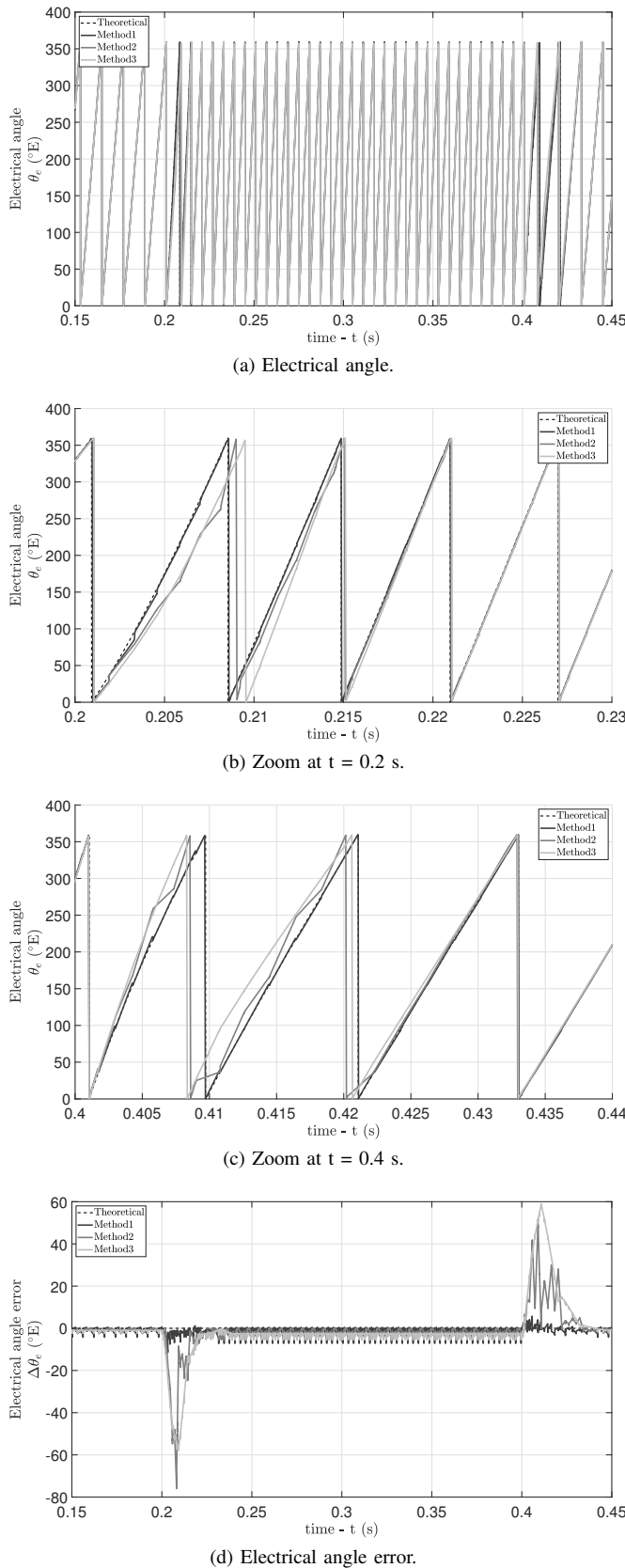


Fig. 10. Estimated electrical angle and electrical angle error during a speed transient if the electric machine runs as a generator.

speed and nominal load rate in terms of RMS and THD phase-current values. Method 2 and the proposed method accomplish

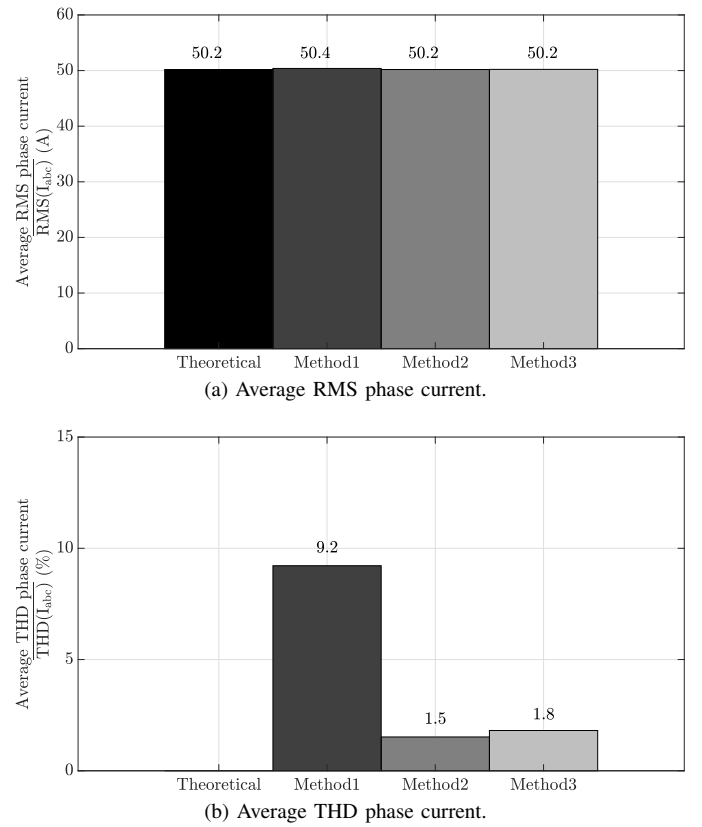


Fig. 11. Simulation results of RMS and THD of phase currents during steady-state if the electric machine runs as a motor at its nominal speed and load.

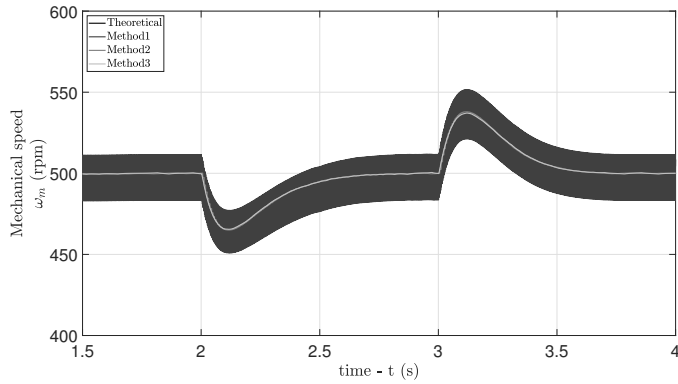
the RMS value closer to the ideal performance, but Method 2 achieves the minimum THD current content that would be caused by the inaccurate representation of the speed and electrical angle.

Fig. 12 shows the transient response of the speed and the electrical angle error when the electric machine runs as a motor. The speed reference is its nominal value. The load increases from 50% to 100% at time 2 s and decreases from 100% to 50% at time 3 s. All methods maintain the speed reference when the load changes while managing the electrical angle error. Furthermore, yet Method 1 estimates the electrical speed with oscillations, the FOT approximation filters its consequences on the electrical angle calculation.

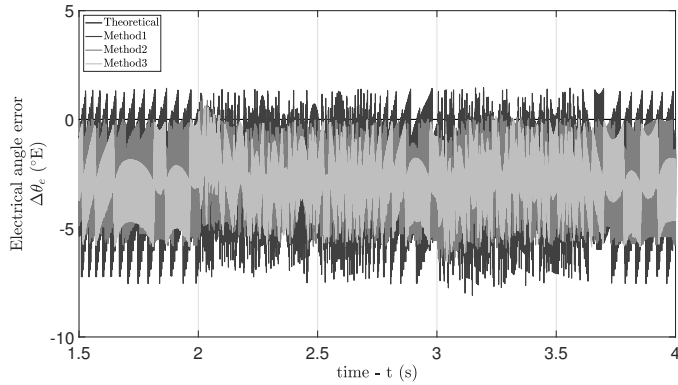
V. EXPERIMENTAL RESULTS

The feasibility of the proposed method is likewise verified through an experimental testbed (see Fig. 13). The testing platform comprises a 6.5 kW three-phase inverter, fed from a rated 48 V lithium-ion battery, and a 2.5 kW and 500 rpm IPMSM. The IPMSM has been tightly coupled to a DC machine through a chain. A TI DSP TMS320F2609 controls the traction inverter. FOC algorithm is executed each 50 μ s at the control loop, and Hall information is processed each 556 μ s at the maximum speed.

Fig. 14 shows the traction inverter performance when running as a generator at different speed rates (50%, 100%, and 150% of its nominal speed). The estimated speed is buffered at 1 ms, 0.5 ms, and 0.25 ms when the electrical



(a) Mechanical speed.



(b) Electrical angle error.

Fig. 12. Estimated mechanical speed and electrical angle error during a load transient if the electric machine runs as a motor at its nominal speed.

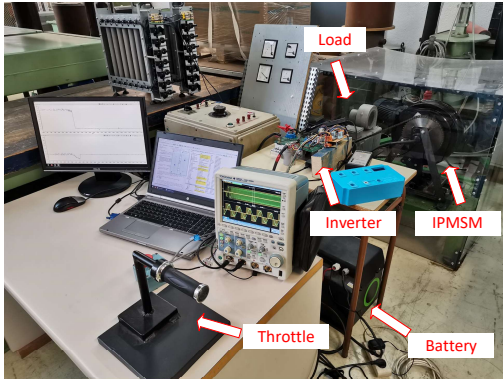
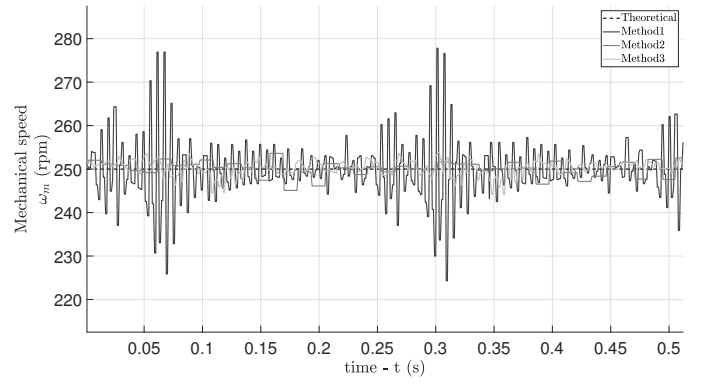


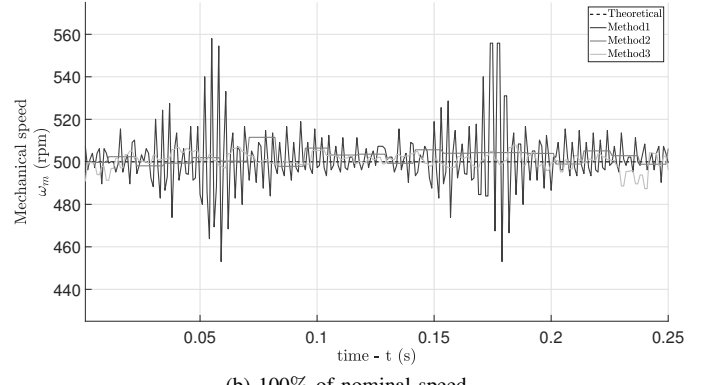
Fig. 13. Experimental setup: battery, traction inverter, IPMSM and load.

machine runs at 50%, 100%, and 150% of its nominal speed, respectively. Method 1 introduces oscillations in the electrical speed estimation because the calculation is based on the time difference between two consecutive edges, and it presents some distortion with misplaced HESSs, as can be observed in Fig. 5. Method 2 is based on the Last Square Method with the last six samples of the previous electrical period, and the linear regression minimizes the HESS deviation. At steady-state, Method 3 makes no error in the electrical speed estimation by using (3) because it is not affected by the misplacement of HESSs, as shown in Fig. 5.

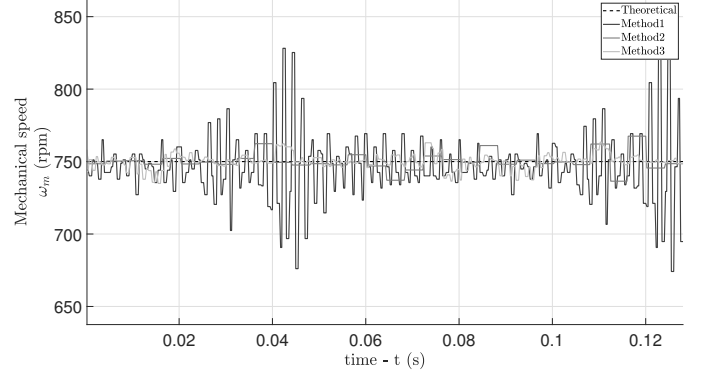
Fig. 15 shows the steady-state performance of the traction inverter when the electric machine runs as a motor at nominal



(a) 50% of nominal speed.



(b) 100% of nominal speed.



(c) 150% of nominal speed.

Fig. 14. Estimated mechanical speed during steady-state if the electric machine runs as a generator at different speed rates.

speed and nominal load rate in terms of RMS and THD phase-current values. Similar to the simulation results, Method 2 and Method 3 consume the lowest RMS values for the same torque setpoint, and the THD content is very similar in both cases.

Table IV summarises the steady-state response comparing the enumerated methods throughout the memory usage and the burden time in an actual microcontroller, the Coefficient of Variation of the Root-Mean Squared Error - CV(RMSD) of the speed estimation as a generator, and the average values of the RMS and THD of the consumed current at nominal speed and nominal load running as a motor. Fig. 16 depicts a radar diagram from the previous results where the proposed method has the minimum area, and it represents the strategy that achieves the best performance under the selected parameters.

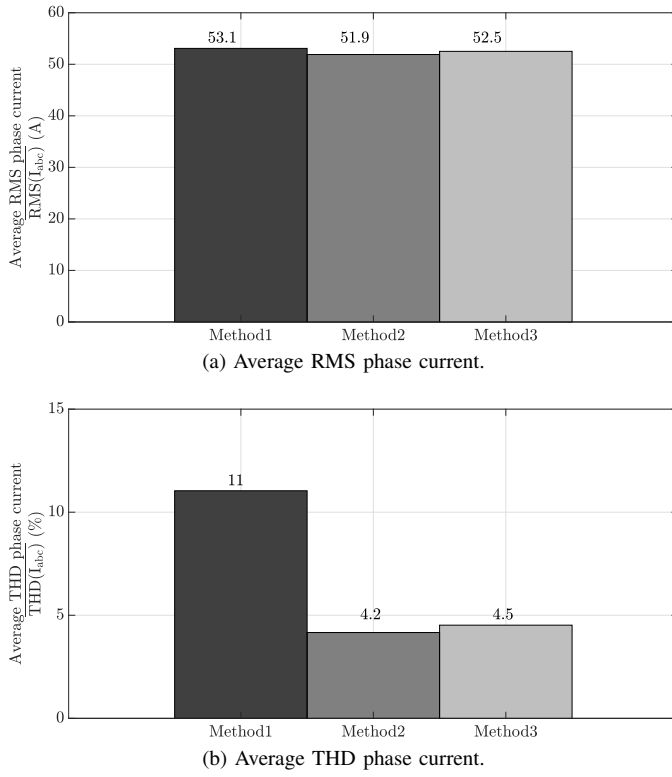


Fig. 15. Experimental results of RMS and THD of phase currents during steady-state if the electric machine runs as a motor at its nominal speed and load.

TABLE IV
EXPERIMENTAL RESULTS COMPARISON

Strategy	Memory (bytes)	Time (cycles)	Speed error (rpm)	Current RMS (A)	Current THD (%)
Method 1	1012	350	3.0	53.1	11.0
Method 2	2186	1314	0.8	51.9	4.2
Method 3	1222	450	0.7	52.5	4.5

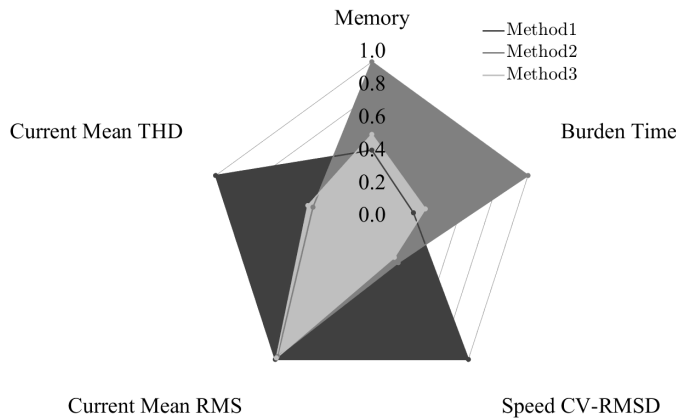


Fig. 16. Per-unit experimental results comparison at steady-state.

Fig. 17 shows the transient response of the traction inverter due to torque reference steps and a speed limitation working in both the MTPA and FW zones in the experimental setup. Torque reference changes from 0 Nm to 40 Nm and from 40 Nm to 80 Nm at time equal to 0 s and 2 s, respectively,

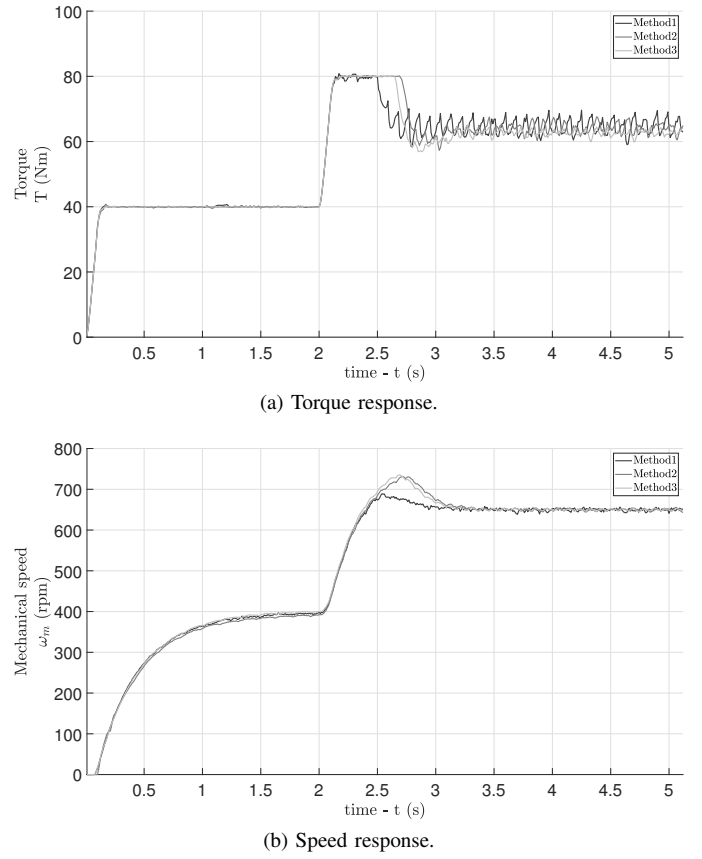


Fig. 17. Estimated torque and speed during different torque reference transients, and limiting the maximum speed, if the electric machine runs as a motor in the experimental test bed.

with a limit speed of 650 rpm. The calculated torque and the estimated speed are buffered at each 10 ms with a total of 512 samples. The calculated torque and the estimated speed are buffered at each 10 ms with a total of 512 samples. The mechanical speed waveform is graphically filtered concerning the results depicted in Fig. 14. Method 1 starts to saturate the torque setpoint before due to the high-frequency oscillations of the speed estimation. However, Method 2 and Method 3 enlarge the maximum torque for more time until the speed limit is achieved due to a better representation of the electrical speed. Method 3 gets the minimum torque ripple during the steady-state.

VI. CONCLUSION

This paper has proposed an accurate method to estimate the speed and electrical angle from misplaced HESSs in PMSMs. Besides, the presented method has been compared with state-of-the-art approaches for similar applications. The main feature of this technique is to minimise the harmonic content of the phase currents due to an inaccurate speed or electrical angle representation during steady-state conditions, along with a quick speed response during transient conditions. When implemented in a real microcontroller, considering the critical points of the comparison memory, burden time, speed estimation error, and output current waveform, the proposed method presents the best performance at steady-state speed

and torque conditions and during transient with speed and load steps. In different test cases, simulation and experimental results certify the methodology efficacy on an existing low-voltage electric powertrain.

REFERENCES

- [1] T. Møller and J. Simlett, "Micromobility: moving cities into a sustainable future," EY Mobility Centre, Tech. Rep., 2020. [Online]. Available: https://assets.ey.com/content/dam/ey-sites/ey-com/en_gl/topics/automotive-and-transportation/automotive-transportation-pdfs/ey-micromobility-moving-cities-into-a-sustainable-future.pdf
- [2] T. Finken, M. Felden, and K. Hameyer, "Comparison and design of different electrical machine types regarding their applicability in hybrid electrical vehicles," *2008 18th International Conference on Electrical Machines*, 2008, pp. 1–5, doi: 10.1109/ICELMACH.2008.4800044.
- [3] B. Gu, J. Choi, and I. Jung, "Development and Analysis of Interturn Short Fault Model of PMSMs With Series and Parallel Winding Connections," in *IEEE Transactions on Power Electronics*, vol. 29, no. 4, pp. 2016–2026, Apr. 2014, doi: 10.1109/TPEL.2013.2265400.
- [4] F. Niu, B. Wang, A. S. Babel, K. Li, and E. G. Strangas, "Comparative Evaluation of Direct Torque Control Strategies for Permanent Magnet Synchronous Machines," in *IEEE Transactions on Power Electronics*, vol. 31, no. 2, pp. 1408–1424, Feb. 2016, doi: 10.1109/TPEL.2015.2421321.
- [5] W. Kim and S. Kim, "A Sensorless V/f Control Technique based on MTPA Operation for PMSMs," *2018 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2018, pp. 1716–1721, doi: 10.1109/ECCE.2018.8557412.
- [6] V. M. Bida, D. V. Samokhvalov, and F. S. Al-Mahturi, "PMSM vector control techniques - A survey," *IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering (E IConRus)*, 2018, pp. 577–581, doi: 10.1109/EIConRus.2018.8317164.
- [7] O. Wallmark, L. Harnefors, and O. Carlson, "An improved speed and position estimator for salient permanent-magnet synchronous motors," in *IEEE Transactions on Industrial Electronics*, vol. 52, no. 1, pp. 255–262, Feb. 2005, doi: 10.1109/TIE.2004.841088.
- [8] G. Wang, Z. Li, G. Zhang, Y. Yu, and D. Xu, "Quadrature PLL-Based High-Order Sliding-Mode Observer for IPMSM Sensorless Control With Online MTPA Control Strategy," in *IEEE Transactions on Energy Conversion*, vol. 28, no. 1, pp. 214–224, Mar. 2013, doi: 10.1109/TEC.2012.2228484.
- [9] Y. Kano and N. Matsui, "Sensorless control of interior permanent magnet synchronous motor: An overview and design study," *2017 IEEE Workshop on Electrical Machines Design, Control and Diagnosis (WEMDCD)*, 2017, pp. 199–207, doi: 10.1109/WEMDCD.2017.7947747.
- [10] X. Song, J. Fang, and B. Han, "High-Precision Rotor Position Detection for High-Speed Surface PMSM Drive Based on Linear Hall-Effect Sensors," in *IEEE Transactions on Power Electronics*, vol. 31, no. 7, pp. 4720–4731, Jul. 2016, doi: 10.1109/TPEL.2015.2479642.
- [11] Q. Tang, A. Shen, G. Luo, and J. Xu, "PMSM Sensorless Control by Injecting HF Pulsating Carrier Signal into ABC Frame," in *IEEE Transactions on Power Electronics*, vol. 32, no. 5, pp. 3767–3776, May 2017, doi: 10.1109/TPEL.2016.2583787.
- [12] P. Kumar, O. Bottesi, S. Calligaro, L. Alberti, and R. Petrella, "Self-Adaptive High-Frequency Injection Based Sensorless Control for Interior Permanent Magnet Synchronous Motor Drives," in *Energies*, vol. 12, no. 3645, pp. 1–26, Sep. 2019, doi: 10.3390/en12193645.
- [13] Y. Yang and Y. Ting, "Improved Angular Displacement Estimation Based on Hall-Effect Sensors for Driving a Brushless Permanent-Magnet Motor," in *IEEE Transactions on Industrial Electronics*, vol. 61, no. 1, pp. 504–511, Jan. 2014, doi: 10.1109/TIE.2013.2247013.
- [14] A. Lidozzi, L. Solero, F. Crescimbeni, and A. Di Napoli, "SVM PMSM Drive With Low Resolution Hall-Effect Sensors," in *IEEE Transactions on Power Electronics*, vol. 22, no. 1, pp. 282–290, Jan. 2007, doi: 10.1109/TPEL.2006.886603.
- [15] T. D. Batzel and K. Y. Lee, "Slotless permanent magnet synchronous motor operation without a high resolution rotor angle sensor," in *IEEE Transactions on Energy Conversion*, vol. 15, no. 4, pp. 366–371, Dec. 2000, doi: 10.1109/60.900494.
- [16] S. Morimoto, K. Kawamoto, M. Sanada, and Y. Takeda, "Sensorless control strategy for salient-pole PMSM based on extended EMF in rotating reference frame," in *IEEE Transactions on Industry Applications*, vol. 38, no. 4, pp. 1054–1061, Jul. 2002, doi: 10.1109/TIA.2002.800777.
- [17] P. Alaeinovin, S. Chiniforoosh, and J. Jatskevich, "Evaluating misalignment of hall sensors in brushless DC motors," *2008 IEEE Canada Electric Power Conference*, 2008, pp. 1–6, doi: 10.1109/EPC.2008.4763350.
- [18] S. Tsotoulidis and A. Safacas, "Side-effects of Hall sensors misplacement on BLDC motor drive operation," *2014 International Conference on Electrical Machines (ICEM)*, 2014, pp. 1825–1830, doi: 10.1109/ICELMACH.2014.6960431.
- [19] Y. Lu, L. Liu, and D. Zhang, "Simulation and Analysis of Thermal Fields of Rotor Multislots for Nonsalient-Pole Motor," in *IEEE Transactions on Industrial Electronics*, vol. 62, no. 12, pp. 7678–7686, Dec. 2015, doi: 10.1109/TIE.2015.2455514.
- [20] S. Morimoto, M. Sanada, and Y. Takeda, "Sinusoidal current drive system of permanent magnet synchronous motor with low resolution position sensor," *IAS '96. Conference Record of the 1996 IEEE Industry Applications Conference Thirty-First IAS Annual Meeting*, 1996, vol. 1, pp. 9–14, doi: 10.1109/ias.1996.556990.
- [21] F. Giulii Capponi, G. De Donato, L. Del Ferraro, O. Honorati, M. C. Harke, and R. D. Lorenz, "AC brushless drive with low-resolution Hall-effect sensors for surface-mounted PM machines," in *IEEE Transactions on Industry Applications*, vol. 42, no. 2, pp. 526–535, Mar. 2006, doi: 10.1109/TIA.2005.863904.
- [22] S. B. Ozturk, B. Akin, H. A. Toliyat, and F. Ashrafzadeh, "Low-cost direct torque control of permanent magnet synchronous motor using hall-effect sensors," *Twenty-First Annual IEEE Applied Power Electronics Conference and Exposition, APEC '06*, 2006, p. 7, doi: 10.1109/APEC.2006.1620610.
- [23] X. Zhang and W. Zhang, "An improved rotor position estimation in PMSM with low-resolution hall-effect sensors," *2014 17th International Conference on Electrical Machines and Systems (ICEMS)*, 2014, pp. 2722–2727, doi: 10.1109/ICEMS.2014.7013961.
- [24] L. Kreindler, I. Iacob, G. Casaru, A. Sarca, R. Olteanu, and D. Matianu, "PMSM drive using digital hall position sensors for light EV applications," *2015 9th International Symposium on Advanced Topics in Electrical Engineering (ATEE)*, 2015, pp. 199–204, doi: 10.1109/ATEE.2015.7133764.
- [25] H. Mehta, V. Joshi, U. Thakar, M. Kuber, and P. Kurulkar, "Speed control of PMSM with Hall sensors using DSP TMS320F2812," *2015 IEEE 11th International Conference on Power Electronics and Drive Systems*, 2015, pp. 295–300, doi: 10.1109/PEDS.2015.7203482.
- [26] A. Yoo, S. Sul, D. Lee, and C. Jun, "Novel Speed and Rotor Position Estimation Strategy Using a Dual Observer for Low-Resolution Position Sensors," in *IEEE Transactions on Power Electronics*, vol. 24, no. 12, pp. 2897–2906, Dec. 2009, doi: 10.1109/TPEL.2009.2022969.
- [27] Z. Wang, K. Wang, J. Zhang, C. Liu, and R. Cao, "Improved rotor position estimation for permanent magnet synchronous machines based on hall-effect sensors," *2016 IEEE International Conference on Aircraft Utility Systems (AUS)*, 2016, pp. 911–916, doi: 10.1109/AUS.2016.7748184.
- [28] S. Kim, C. Choi, K. Lee, and W. Lee, "An Improved Rotor Position Estimation With Vector-Tracking Observer in PMSM Drives With Low-Resolution Hall-Effect Sensors," in *IEEE Transactions on Industrial Electronics*, vol. 58, no. 9, pp. 4078–4086, Sep. 2011, doi: 10.1109/TIE.2010.2098367.
- [29] J. A. Cortajarena, S. García, J. Cortajarena, O. Barambones, and P. Alkorta, "Influence of the rotor angle precision in control of interior permanent magnet synchronous machine drives and improvement method using sensorless estimator with Hall sensors," in *IET Power Electronics*, vol. 12, no. 3, pp. 383–391, Jan. 2019, doi: 10.1049/iet-pel.2018.5143.
- [30] L. Yu, Y. Zhang, and W. Huang, "Accurate and Efficient Torque Control of an Interior Permanent Magnet Synchronous Motor in Electric Vehicles Based on Hall-Effect Sensors," in *Energies*, vol. 10, no. 3, p. 410, Mar. 2017, doi: 10.3390/en10030410.
- [31] Y. Zhao, W. Huang, J. Yang, F. Bu, and S. Liu, "A PMSM rotor position estimation with low-cost Hall-effect sensors using improved PLL," *2016 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC Asia-Pacific)*, 2016, pp. 804–807, doi: 10.1109/ITEC-AP.2016.7513058.
- [32] G. Liu, B. Chen, and X. Song, "High-Precision Speed and Position Estimation Based on Hall Vector Frequency Tracking for PMSM with Bipolar Hall-Effect Sensors," in *IEEE Sensors Journal*, vol. 19, no. 6, pp. 2347–2355, Mar. 2019, doi: 10.1109/ISEN.2018.2885020.
- [33] C. Miguel-Espinar, D. Heredero-Peris, G. Igor-Gross, M. Llonch-Masachs, and D. Montesinos-Miracle, "An enhanced electrical angle representation in PMSM control with misplaced Hall-effect switch sensors," *2020 23rd International Conference on Electri-*

cal Machines and Systems (ICEMS), 2020, pp. 1454–1459, doi: 10.23919/ICEMS50442.2020.9291194.

- [34] C. Miguel-Espinar, D. Heredero-Peris, G. Gross, M. Llonch-Masachs, and D. Montesinos-Miracle, “Maximum Torque per Voltage Flux-Weakening Strategy With Speed Limiter for PMSM Drives,” in *IEEE Transactions on Industrial Electronics*, vol. 68, no. 10, pp. 9254–9264, Oct. 2021, doi: 10.1109/TIE.2020.3020029.



Carlos Miguel-Espinar was born in L'Hospitalet de Llobregat, Spain, in 1985. He obtained the MSc. degree in Electrical Engineering from the ETSEIB, at the UPC, in 2011. Since 2011 he has been working at the CITCEA-UPC as a senior researcher developing R&D projects for the industry in various areas: electronics, power electronics, digital control and communications, electric machine design, induction heating, and others. He is currently a PhD candidate at the UPC to develop advanced control algorithms for PMSMs.



Daniel Heredero-Peris was born in Vilanova i la Geltrú, Spain, in 1985. He obtained the MSc. degree in Control Engineering from the ETSEIB, UPC, in 2010, and PhD degree from the UPC, in 2017. Since 2010 he is working at CITCEA-UPC as a project engineer developing tasks focused on the design of algorithms related to the control of grid-connected and standalone single/three-phase inverters, and V2G projects for EV. His main research interests are control of power electronics, micro-grids and EVs.



Gabriel Gross was born in Ibiza, Spain, in 1979. He received his MSc. degree in Electrical Engineering from the ETSEIB, at the UPC in 2004. Since then he is working at CITCEA-UPC developing R&D projects for the industry in various areas: electronics, power electronics, digital control and communications, electric machine design, and others. First as a project engineer (2005-2008), then as project manager (2009-2011) and later as team manager (2012-now). His primary research interest are power electronics and digital control, focused on electric

motor control.



Marc Llonch Masachs was born in Terrassa, Spain, in 1978. He received the MSc. degree in Industrial Engineering from the ETSEIB, at UPC in 2014. Since 2014 he is working with CITCEA-UPC as a project engineer carrying out tasks focused on the development of grid connected and standalone converters for DERs and EVs. His main research interest is focused in the power electronic knowledge at the service of micro-grids and EVs.



Daniel Montesinos-Miracle (S'01-M'08-SM'12) was born in Barcelona, Spain, in 1975. He obtained the MSc. degree in Electrical Engineering from the ETSEIB, UPC, in 2000, and PhD degree from the UPC, in 2008. In 2001 he joined Salicru Electronics S.A., Santa Maria de Palautordera, as a R&D engineer. Since 2001 he has been involved in CITCEA-UPC as a research collaborator. In 2005 he became a lecturer at Electrical Engineering Department, UPC. Since 2012 he has been an Associate Professor at UPC. He has become the CITCEA-UPC director in

2016. His primary research interests are power electronics, drives and green energy converters.